

Northern Territory US Defence Engagement Strategy

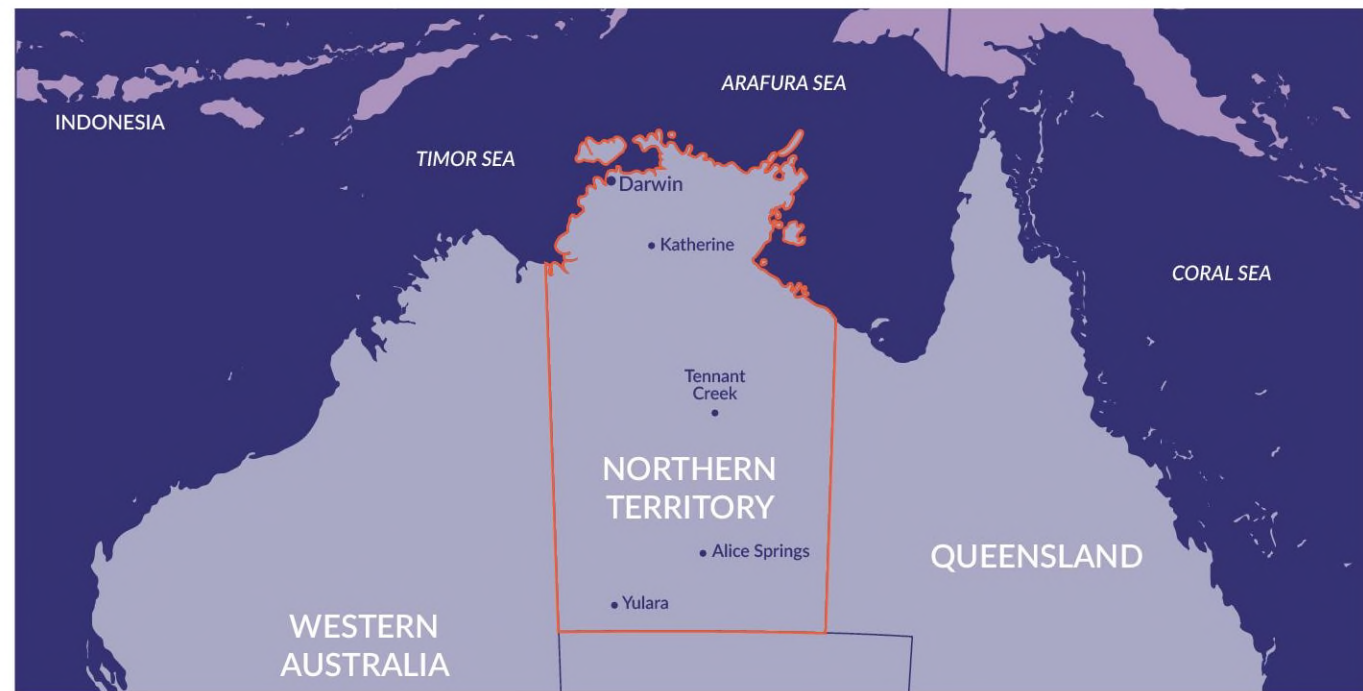


Deepening bilateral defence ties, set against the ground-breaking AUKUS agreement and Quadrilateral leaders meeting in September 2021, are accelerating the transformation of the Northern Territory into a vital alliance hub.

It is now accepted that previous levels of defence cooperation and interoperability are no longer sufficient to meet the challenges ahead. The US and Australia need to pool their resources, jointly fund critical infrastructure, collaborate across all areas of defence technology,

step-up training and exercises and become super-interoperable. The Territory offers unrivalled opportunities to achieve these goals and to become a potent force multiplier for the US and Australian militaries.





The strategic importance of Australia and the Northern Territory

Rising great power competition in the Indo-Pacific is a timely reminder of the importance of the Northern Territory to the defence of Australian and American interests in the region.

As President Joe Biden moves to strengthen and reshape the US military to meet the China challenge Washington views Australia as a principal ally and partner. For the first time since World War II, American and Australian areas of strategic priority overlap in the Indo-Pacific. More than a decade ago President Barack Obama characterized the alliance with Australia as “indispensable”, a view shared today by senior officials and military leaders in the US Department of Defense, most members of Congress and influential members of the Biden administration.

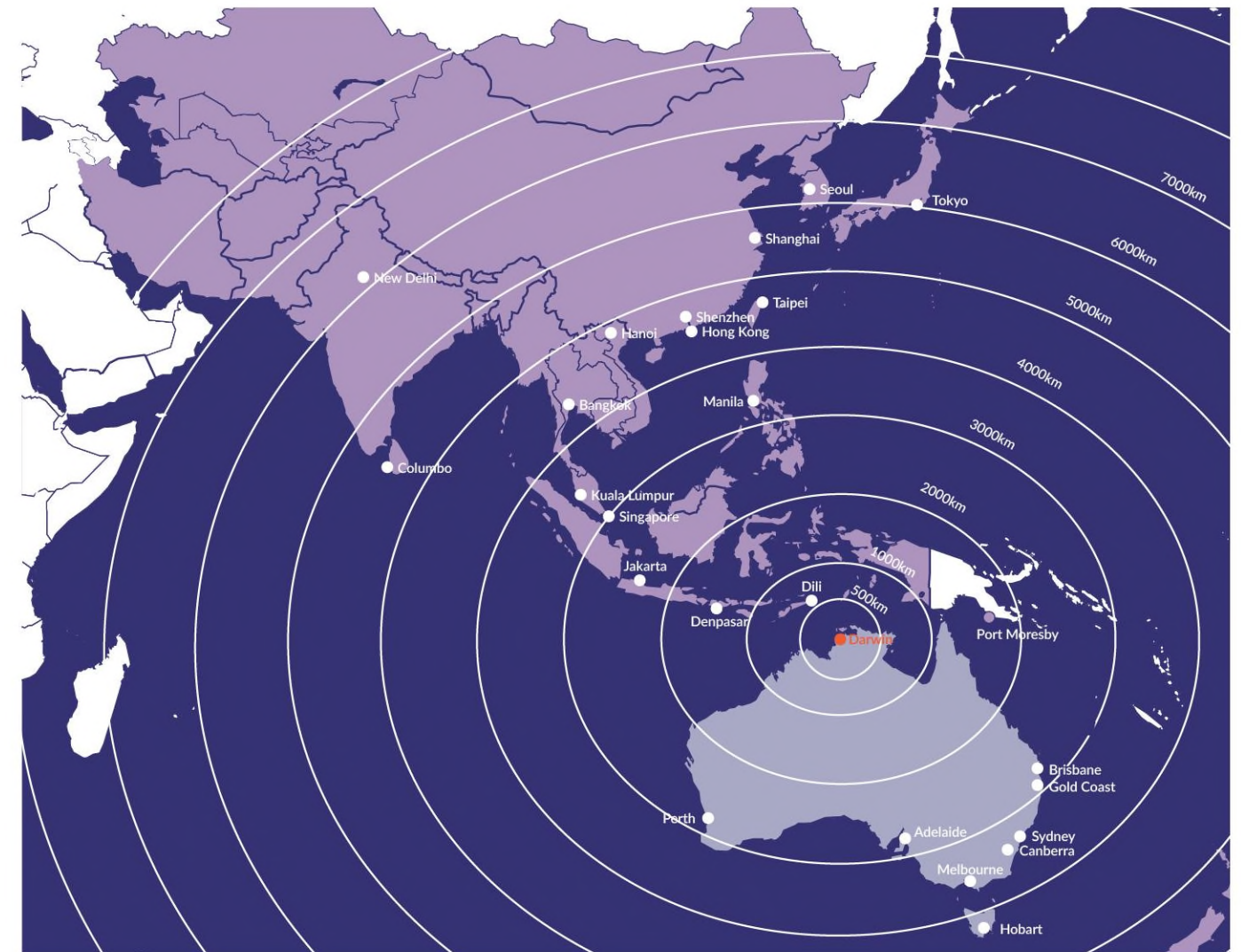
This reflects the strong ties that have historically bound the two democracies and Australia’s proven record as a trusted provider of security goods. The Territory is uniquely placed to support alliance operations due to its pivotal location at the maritime cross-roads of the Pacific and Indian Oceans. The South China Sea and Malacca Strait carry around 50 percent of global energy exports, 40 percent of sea-borne trade and 80 percent of China’s oil imports.

The spread of destabilizing military technologies, hybrid warfare and Beijing’s determination to establish a China-centric, authoritarian international order, have seriously degraded the regional security environment. At this inflexion point in the region’s development and security, the Northern Territory is emerging as a determining enabler of alliance strategy.

The Northern Territory

A crucial allied bastion during the Second World War, the northern port city of Darwin is once again central to Australian and US defence planning. The capital of the Northern Territory – a sparsely populated but resource rich jurisdiction twice the size of Texas – Darwin is the main forward operating base for the Australian Defence Force. With further investment, Australia’s “Top End” could become a major mounting and support base, training destination, experimentation hub and force multiplier for the US and other allies and partners.

The Territory offers solutions to many of the problems confronting the US military in responding to the China challenge and other pressing security concerns in the Indo-Pacific. These solutions stem from the Territory’s proximity to Southeast Asia; favourable geography for multi-domain operations; potential for low-cost force structure; established position as an alliance hub; possessor of a deep-water port, supporting marine precinct and world class training ranges and exercise areas; developing aerospace sector; transformative digital infrastructure; and ability to produce high value rare earths for US defence industry.



Geography matters

The Territory occupies a geographical “sweet spot” in the Indo-Pacific offering easy access and rapid force deployment to the Western Pacific and Eastern Indian Oceans. 3340 kms from Singapore, Darwin is close to potential conflict “hot spots” in the Malacca Strait and contested South China Sea. Guam (4700 kms) and Hawaii (11,000 kms) are much further away. Darwin is only 3,000 kms from the Chinese garrison on Mischief Reef in the central South China Sea and relatively close to the strategically located airfield on Australian owned Cocos Island (3686 kms) in the Eastern Indian Ocean. Cocos Island is even closer to the South China Sea and less than 1,200 kms from the Sunda Strait, a strategic choke point.

Aside from its role as a forward operating base for the ADF, the Territory is tailor-made to be an American support and mounting location and Indo-Pacific gatekeeper, enhancing the ability of US forces to control and keep watch over the South China Sea, the world’s most important waterway.

With ample strategic depth, the Territory is a much safer place to operate from than small, island bases where dispersal of defence assets is virtually impossible and logistical support difficult. Beyond the range of most Chinese missiles and strike aircraft, the Territory is far enough away from direct Chinese influence but close enough to affect Chinese actions in the region. It’s also an ideal launching point for an inside force that can react to aggression and not have to fight its way into the conflict.

For these reasons, the Territory is an attractive location for rotating and transiting US ships, aircraft and land forces which are overly reliant on a handful of island bases in the Pacific and Indian Oceans. Guam is already within range of Chinese ballistic missiles and submarine launched cruise missiles. Although more secure, aircraft based on Diego Garcia would take nearly twice as long to reach Mischief Reef as those based in Darwin.

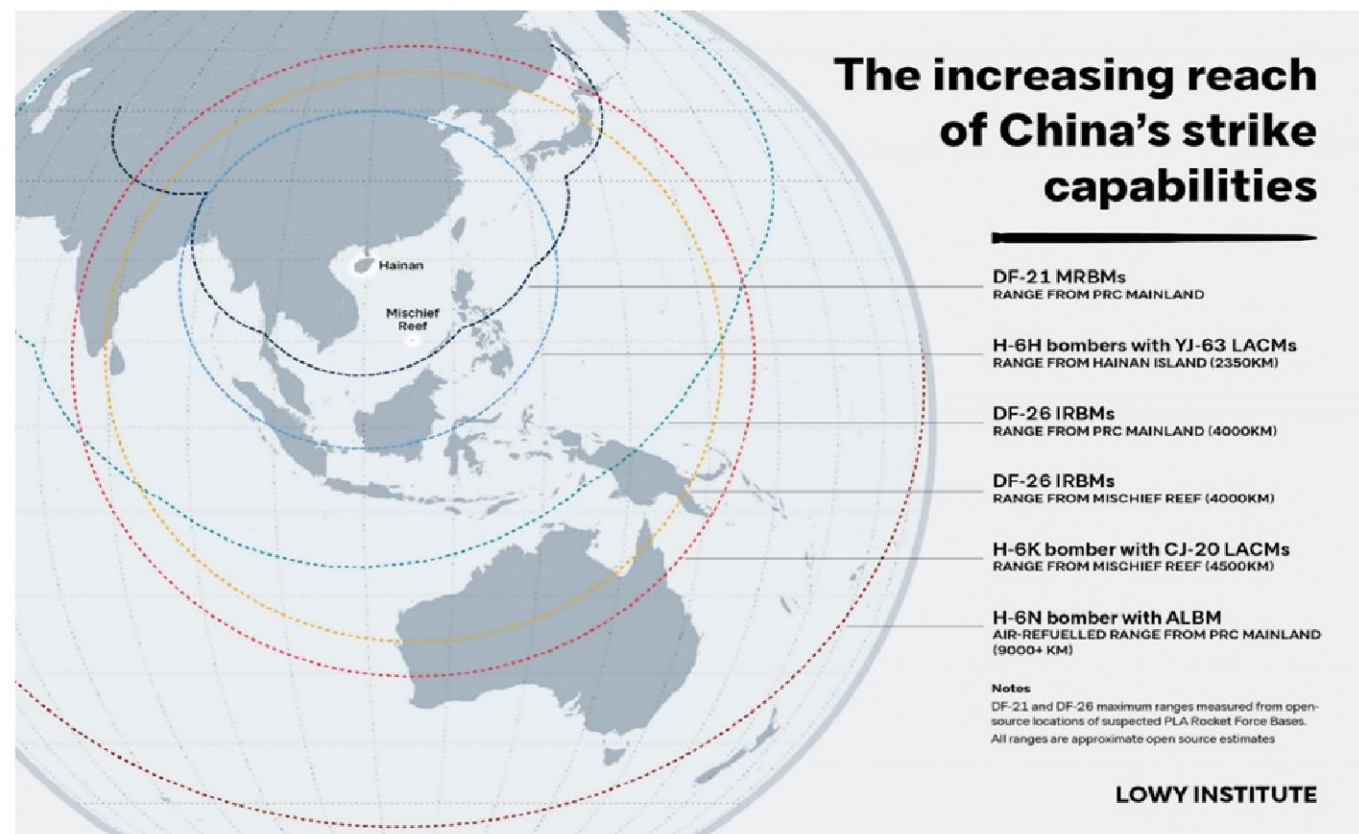


Figure: Chinese Missile and Bomber Ranges

...and so does low-cost force structure

The Territory is well placed to become a significant piece of low-cost force posture for the US military which is under-weight in Asia and poorly configured for the challenge of defending American interests in the wider Indo-Pacific. Unless addressed, the regional capability gap with a rapidly modernizing Peoples Liberation Army will almost certainly widen.

The US is shifting towards a more dispersed force posture in response to China's ability to strike forward deployed US forces. But additional posture won't come quickly easily or cheaply in Southeast Asia. Okinawa is maxed out. And there are political and strategic constraints on what can be done in Japan and South Korea. The Territory is an appealing alternative. It has a growing capacity to host, fuel, replenish and repair visiting ships and aircraft; offers unsurpassed opportunities for training and exercising sophisticated war-fighting skills; has secure, high-speed communications; a capable, scalable industrial base; the capacity to launch satellites and high-altitude balloons; and a defence friendly government and social environment.

The Territory is also positioned to host and support the expected increase in US military assets deployed to Australia for training, exercises and joint operations. Acknowledging that it had been 10 years since the establishment of the United States Force Posture Initiatives (USFPI) in Australia, Washington and Canberra agreed to intensify force posture cooperation at the September AUSMIN meeting. Re-established at AUSMIN 2020, the bilateral Force Posture Working Group convened in

May 2021 to develop recommendations to promote a secure and stable Indo-Pacific region and deter adversaries. AUSMIN has agreed to:

- Enhance air cooperation through the rotational deployment of US aircraft of all types in Australia and appropriate aircraft training and exercises.
- Enhance maritime cooperation by increasing logistics and sustainment capabilities of US surface and subsurface vessels in Australia.
- Enhance land cooperation by conducting more complex and more integrated exercises and greater combined engagement with allies and partners in the region.
- Establish a combined logistics, sustainment and maintenance enterprise to support high end warfighting and combined military operations in the region.

In a powerful reaffirmation of its commitment to the alliance, Australia will work with the US and UK to acquire eight nuclear-powered submarines, a suite of long-range missiles and to manufacture munitions and guided missiles under licence from the US. The Territory government is working to make Darwin Harbour a nuclear capable port able to facilitate visits of nuclear-powered ships and submarines and allow them to operate from Darwin, along with supporting fleets of autonomous uninhabited underwater vehicles.

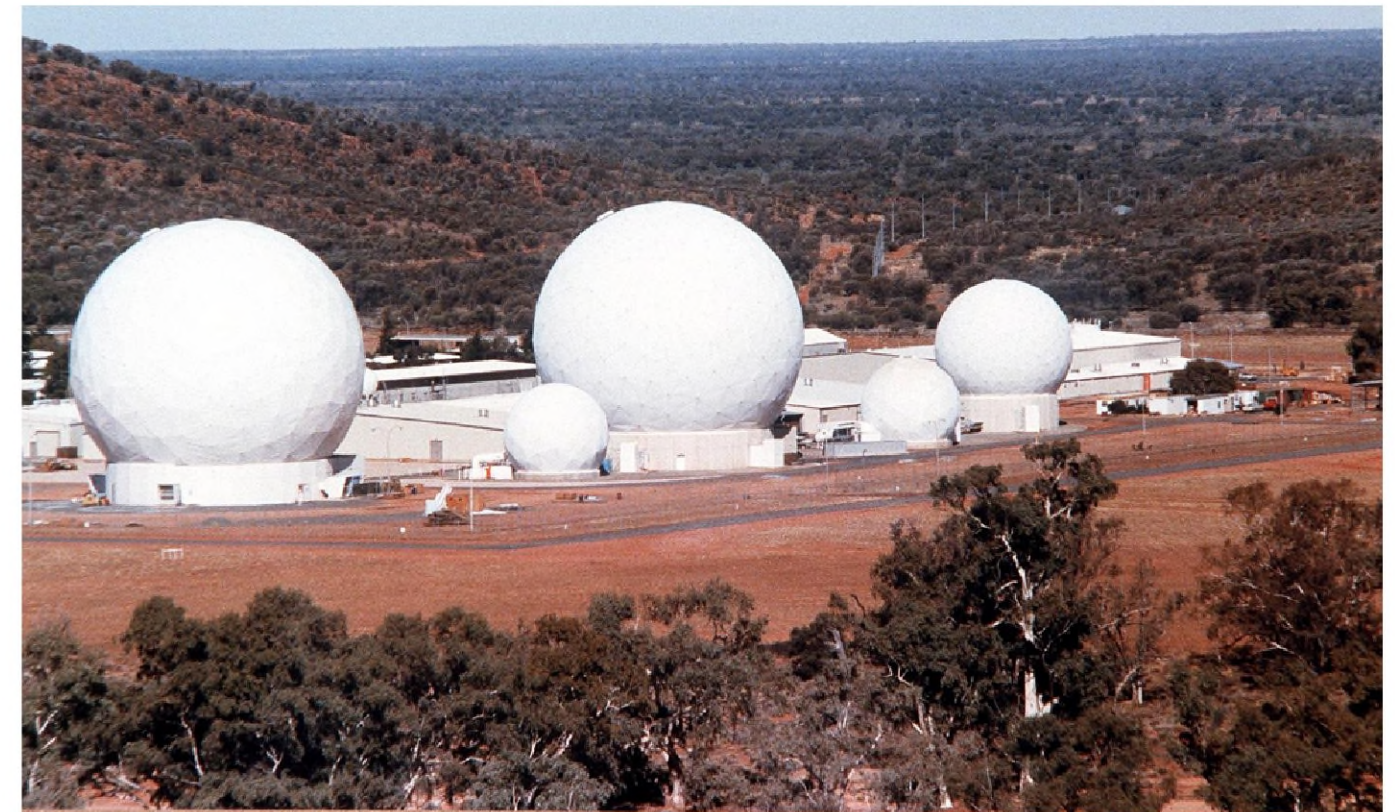


Figure: Pine Gap Alice Springs

Established alliance hub

The Territory could become the linchpin in the chain of US bases and places running from the Aleutians in the North Pacific to Diego Garcia in the Indian Ocean which constrain China and support US strategy and military operations.

There are already several shared US and Australian defence assets in the Territory underpinned by a significant number of in-place bilateral agreements and established cooperative operational and training events. The Joint Defence Facility at Pine Gap near Alice Springs, 1500 kilometres south of Darwin, is the most important intelligence collection, targeting and ballistic missile early warning facility outside the continental United States.

Elements of a Marine Expeditionary Force have rotated through the Territory for a decade training and exercising with the ADF and other US and partner nation militaries. There are regular ship and aircraft visits; a strategic fuel reserve is being built outside Darwin to support future US deployments; a modest level of infrastructure for maintenance and sustainment of military equipment between exercises is in place and more is under consideration as the US adjusts its defence posture in the Indo-Pacific.

But there is a need and appetite to do more because of the heightened risk of conflict in the region and the long lead times required to develop fit-for-purpose facilities. Chinese "grey zone" tactics designed to undermine our will and attrite our capabilities without triggering a military response will reduce the time to respond in the event of a serious conflict. Sustained investment in the Territory's defence infrastructure would help the US to establish a credible deterrent to hostile powers by addressing grey zone tactics while reassuring developing Asian states of the US commitment to a peaceful and prosperous Indo-Pacific.

To mitigate the risk of pre-emptive strikes on existing US bases, facilities and ISR assets in Japan, South Korea and Guam the Territory could provide logistical, combat and intelligence support to US forces deployed in the region reducing some of the capability gaps identified in the Pacific Deterrence Initiative. They include expeditionary airfield and port infrastructure; fuel and munitions storage; maintenance of deployed assets; staging and mounting areas; supply depots; and testing ranges and exercise areas for allies and partners. Darwin is well placed to become a Combined Maritime Forces Headquarters for the alliance, including a reconstituted US First Fleet should Washington decide to proceed with the initiative.



Figure: HMAS Canberra undergoing maintenance in Darwin after Exercise Talisman Sabre

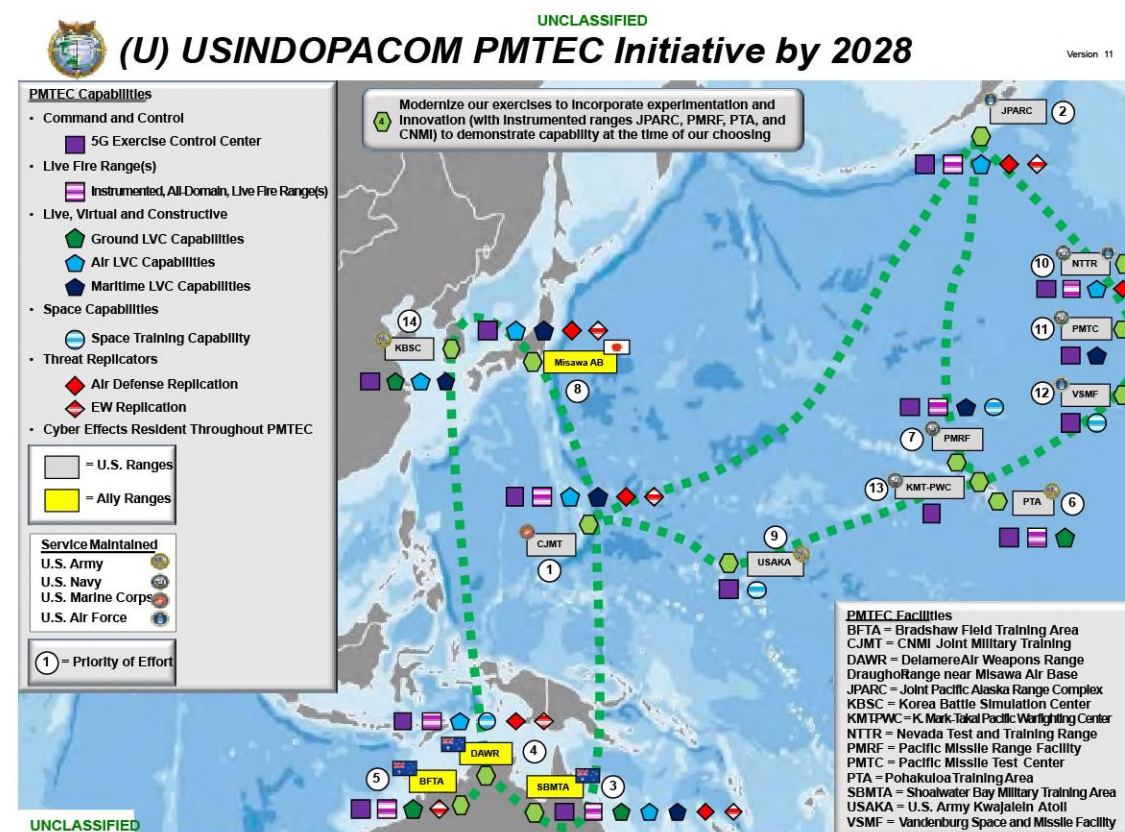
Darwin Harbour and marine precinct: A valuable piece of alliance real-estate

Five times larger than Sydney Harbour and deeper than San Francisco Bay, Darwin Harbour is a valuable piece of alliance real estate with significant potential for expansion and development. This is due to its strategic location, abundant space, relative security from missile attack, proximity to two major defence airfields and a rapidly developing maritime precinct at East Arm adjacent to the existing port. Over the next five years, much of the harbour's growth will be in the defence and national security domains as the Australian and allied navies ramp up their capacity to protect vital sealines of communication and strategic choke points.

Darwin Harbour has the capacity to become a valuable joint-user facility for the US Navy, Marine Corps and fellow Quadripartite members Japan and India, as well as forward deployed NATO forces. The Australian Navy's new focus on the Territory as an operational and sustainment base to support naval operations to Australia's north opens the door to additional Australian defence investment.

The construction of a 5,000-tonne ship lift and the establishment of a regional maintenance centre in Darwin under the Australian Navy's Project Galileo will complement, and improve, the Territory's ability to support and maintain alliance fleets. An increase in the size of the ship lift would allow the speedy repair of battle-damaged destroyers and frigates. In any future contingency, larger vessels could potentially be accommodated in a floating dock moored in the vicinity of the marine precinct which would ideally include a ship repair wharf.

Work is already well underway to provide new wharf space and infrastructure at Darwin Naval Base. Most of the Australian Army's new amphibious watercraft will be located in Darwin following the government's decision to invest up to \$800 million to upgrade the Army's amphibious capabilities. And the Australian Navy plans to homeport a sizeable number of its future mine countermeasures vessels in Darwin, which could contribute towards a multi-national force to keep regional choke points free of sea mines.



World class training and exercise facilities

Taken as a whole, the Territory training range complex is superior to anything in the US and region. It is a unique facility of unmatched size and airspace with the additional advantages of proximity to the sea and virtually no existing electromagnetic signature. The Bradshaw Field Training Area and Delamere Electronic Warfare Range comprise the heart of this complex. Together, they form one of the largest training spaces in the world. With further investment, Darwin could become the defence and national security training provider of choice for the US and region exploiting the Territory's proximity to Asia, abundant natural resources and sparsely populated land mass.

The Territory is already a modest defence training provider. The Australian government has committed to a major \$747 million upgrade programme across all ranges which is expected to be completed by 2026/7 as part of an \$8 billion investment by Canberra in the Territory's capital infrastructure. This includes an overhaul of weapons firing ranges, lengthening the Bradshaw airstrip to support heavier aircraft and new training facilities for ADF personnel and US Marines.

If resourced to a high standard, the US could become a regular user, along with Japan, India, South Korea,

Indonesia, Singapore and Taiwan. The complex could also become an important training location for forward deployed or visiting NATO forces operating in the Indo-Pacific. Digitally connecting these ranges into the USINDOPACOM training network will improve the ability of allies and partners to conduct multi-national operations in all five domains: air, land, sea, space, and cyberspace.

Sophisticated, multinational defence exercises are the most advanced form of training and a logical extension of the Territory's training hub. The Territory has hosted exercises of this nature in the past, notably KAKADU (Australia's largest maritime exercise); Pitch Black (an Air Force oriented multi-national large force deployment); and Talisman Sabre, the main Australian military training exercise with the US and other regional and NATO partners.

Upgrading the range facilities would help make the Territory the preferred destination for such exercises as well as training in complex humanitarian and disaster relief operations, particularly those conducted during high intensity and counter-insurgency combat operations. Plans are underway to add an unmanned air (UAS) and counter-unmanned air (C-UAS) drone "school-house" to the mix and upgrade the 1561 metre runway at Batchelor airfield outside Darwin.



The Upgrades

Robertson Barracks Close Training Area

- Firing ranges
- Weapon training simulation system
- Combat shooting range
- Urban assault range

Kangaroo Flats Training Area

- Marksman's range
- Rifle and grenade assault range
- Demolitions range

Mount Bunday Training Area

- Refurbish training camp
- Ammunition transfer point
- Urban operations facilities
- Aviation facilities

Bradshaw Field Training Area

- Firing range control area
- Medical precinct
- Upgrade maintenance area
- Road upgrades



Figure: Australian Triton

Expanding aerospace complex

The Territory has a small, but expandable aerospace sector with military grade airfields at Darwin, Tindal and near the Arnhem Space Centre at Nhulunbuy. The Territory is recognised as the ideal location for satellite, rocket and balloon launches because of its proximity to the equator and favourable high-altitude winds. The Territory supports the F-35 program and other defence aerospace elements, will soon be home to MQ-4C Triton unmanned aircraft operating out of Tindal and is expected to have the capacity to launch rockets and satellites into Low Earth Orbit from Nhulunbuy by 2022. These capabilities will facilitate intelligence, surveillance and reconnaissance support for allied forces on operations in the Western Pacific and multi-national exercises in the Arafura and Timor Seas to the north of Darwin.

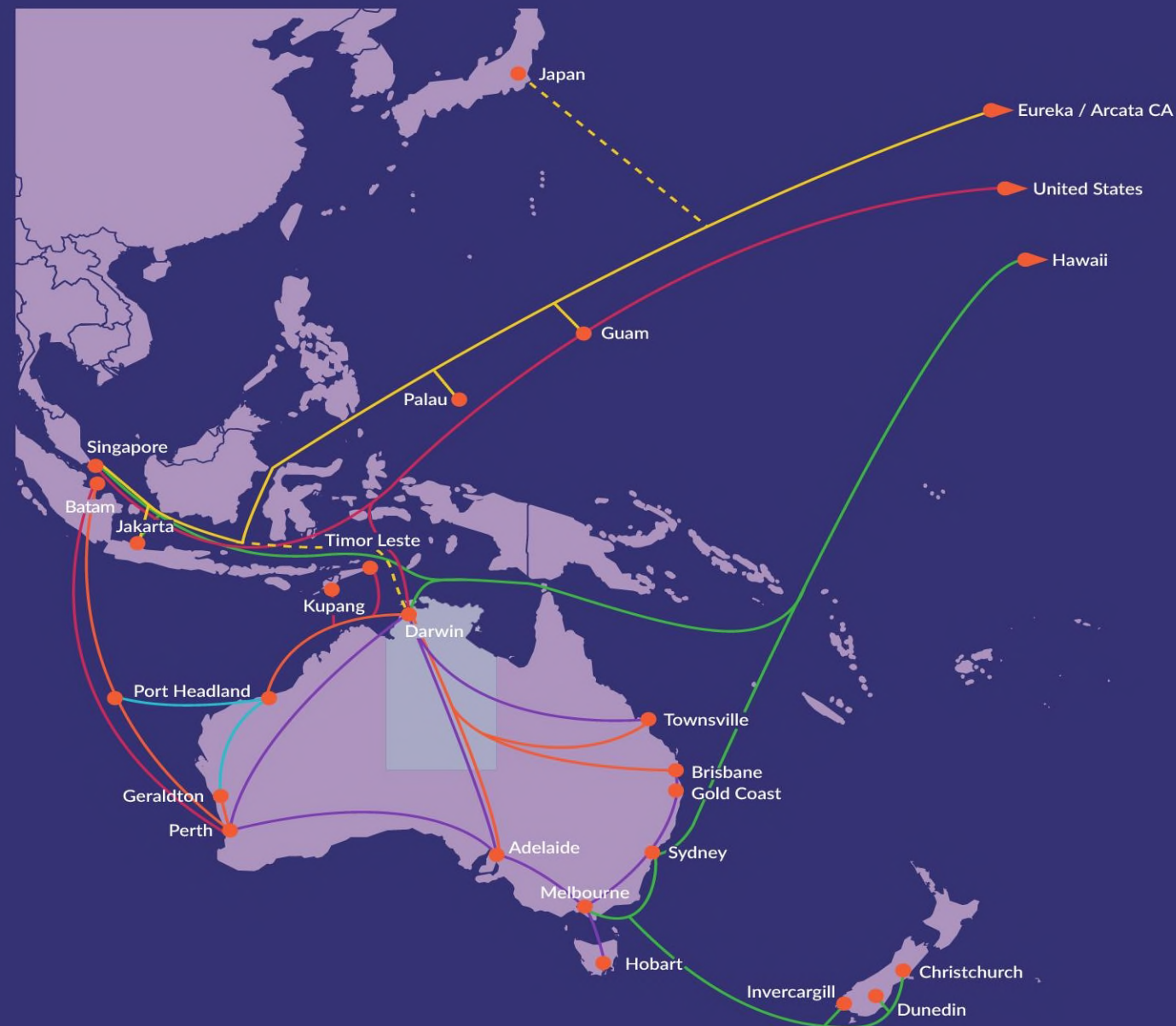
Alice Springs has the potential to become a prime location for the production and launch of stratospheric balloons, which sail the stratospheric winds above the ceiling of conventional aircraft but below space where satellites orbit. Alice Springs' unique advantage for this potentially game changing technology is its proximity to the most favourable stratospheric winds in the world for launching high altitude balloons. These balloons are of increasing interest to the US military because they offer a less expensive layer of communications, high speed digital connectivity and long-range surveillance especially for the location of targets deep inside an adversary's defences.

Both countries recognize the importance of establishing shared capabilities in Space Domain Awareness, Space Command and Control, Satellite Communications and Positioning, Navigation and Timing. The September AUSMIN joint statement reaffirmed the need to develop a common understanding of space-related threats and collaborate over a range of capabilities to leverage the resources, geography and expertise needed to meet the challenges posed by the current strategic environment and achieve alliance objectives.

AUSMIN canvassed plans for a government-to-government Space Framework Agreement that will pave the way for increased cooperation on civil research, exploration and the use of space for peaceful purposes. Also discussed were the ongoing negotiation of a Technology Safeguards Agreement relevant to space activities and a 10-year extension of the NASA-Commonwealth Scientific and Industrial Research Organization (CSIRO) Balloon Flights Agreement governing the launch of high-altitude balloon flights from Australia with instruments regarded as essential to the development of new technologies and payloads for NASA's space flight missions.

These agreements and discussions are strong indicators that Canberra and Washington are on the threshold of a major step-up in aerospace cooperation which plays to the Territory's natural advantages in geography and abundance of emission/people-free land, sea and airspace.





Northern Territory Terrestrial and Sub-Sea Network

Project Echo:

Installed Network and Confirmed Network Extensions:

HyperOne's Network:

Confirmed Extensions to the Vocus Network:

Hawaii Net Cable System:

Planned Network Extension:



Route optimization is also vital for national security networks

Providing a direct route between Singapore and the USA, with branching units to Japan, Guam, Palau, Darwin and Jakarta.

Given today's geopolitical situation, linking the "5 eyes" together with Japan and Singapore using a subsea system that avoids the South China Sea and puts Darwin in a central position.

Transformative digital infrastructure

The Territory government has invested in a major expansion of its digital network, with two international sub-sea connections announced in 2021 and a decision the Territory to a high speed, monitored, sub-sea fibre optic cable running from the US West Coast across the Pacific to Singapore known as Project Echo.

The cable deliberately avoids the contested South China Sea, will increase national security resilience and is expected to be operational in 2023.

Along with the associated data centers that will accompany this investment, Project Echo will provide the redundant, protected, high-speed communications needed to support the vast amount of data processing required to turn information into refined intelligence.

Echo will also enable advanced live, virtual and constructive training for the latest air, land and sea-based platforms to include state of the art aircraft, unmanned ground and aerial systems and ground combat vehicles.



Figure: Rare Earth Magnets

Defence Tech-metal and Renewable Energy Provider

So called “tech metals”, derived from critical minerals, are essential to the global economy and a wide range of weapon systems. The key to their salience is that there are no readily available substitutes. Securing supplies of critical minerals has become a strategic issue for Australia and the US because of their importance to the new economy and China’s dominance of the rare earth’s market. This virtual Chinese monopoly has put rare earths and other high-performance metals on the national security agendas of the US and Australia, as well as many other states.

Washington and Canberra have agreed to improve the security of critical minerals by collaborating on a Critical Minerals Plan of Action “to support new investment in the sector, to diversify supply chains and to consider work with like-minded partners to develop international standards on critical minerals.” In March 2021, a bipartisan group of members of the US House Armed Services Committee formed the Defense Critical Supply Chain Task Force to address perceived US overreliance on vulnerable defence supplies manufactured in other countries. The Report of the Defense Critical Supply Chain Task Force (house.gov) tabled in July, made three main recommendations:

1. That the US government make supply chain security a priority.
2. Do everything it can to protect and strengthen the supply chain for rare earth materials.
3. Work through the National Technology and Industrial Base (which has been expanded to include national security researchers and manufacturers in the US, Canada, the UK, and Australia) to help coordinate efforts with allies to reduce reliance on potentially hostile governments.

The Northern Territory is well endowed with critical minerals. It has 14 of the 17 critical minerals required for electric vehicles and 5th generation platforms and is therefore ideally placed to help the US reduce its supply chain vulnerabilities. The Territory also has substantial commercial reserves of lithium, which are essential for making batteries. Arafura’s Nolans mine, northwest of Alice Springs, contains some of the highest grades of defence relevant neodymium and praseodymium in the world and has already met the Territory’s world leading environmental standards.



Figure: Suncable to supply NT and Singapore with Renewable Energy

With a plenitude of rare earths and established mining expertise, the Territory is well placed to produce the high-performance metals essential for advanced military equipment and capabilities such as the F-35 Joint Strike Fighter; directed energy weapons; laser targeted weapons; smart bombs; drones and cruise missiles, all of which are high priorities for the US.

As the US and allied militaries transition away from reliance on fossil fuels and conventional batteries, renewables and other forms of green energy will become increasingly important for transport and the generation of low-emission electricity and power. The Northern Territory is host to one of the most ambitious solar energy projects in the world. An extensive array of solar panels is being built by Suncable, an Australian founded world-leading energy company, at Elliott half-way between Darwin and Alice Springs.

The Suncable project will harness the Territory’s abundant sunshine to power the Indo-Pacific region with 14GW of competitively priced, dispatchable, high-volume renewable electricity. This solar energy infrastructure network will connect to the Territory’s electricity grid in 2026 and provide 15 percent of Singapore’s electricity via a 3750km subsea HVDC cable system by 2027. The Territory government is currently assessing future defence renewable energy demand to help military users meet their emission targets and power requirements while training and operating from the Territory.



In summary

The Northern Territory's proximity to the locus of great power competition in the most dynamic and contested region of the world provides significant opportunities for low-cost, high-value basing and investment options for the US and other allies and partners. Australia's increasingly tight enmeshment within the alliance, shared strategic interests, highly capable defence force and long history of operating successfully with the US military warrant careful consideration of the advantages the Territory offers for future US force posture, training, mounting, force projection and combat support and sustainment.

Initial investments should be designed to scale-up the Territory's industrial base and defence infrastructure to a level where the US and allies can surge for high intensity operations and training.

Maximising the Territory's offerings will require a collaborative and strategic approach to investment and planning in partnership with the Territory government, Canberra and other partner countries. A strategic partnership of this kind could provide a template for other Australian jurisdictions to help build resilient, networked defence capabilities across northern Australia that would greatly strengthen the US position in the Indo-Pacific.

